

Fall Prevention & Protection Standard

CLEVELAND-CLIFFS STEEL
Fall Prevention & Protection Standard

Effective Date:	<u>11-3-2023</u>	Supersedes:	<u>All Previous Versions</u>
		Standard No.:	<u>CCS_HSS_STD_005.2</u>
		Follow-Up Review Date:	<u>11-3-2024</u>

1 Purpose/Scope..... 2

2 Responsibilities and Authorities..... 3

3 References and Definitions 5

4 Resources10

5 Requirements.....11

6 Forms and Documented Information.....27

7 Opportunities and Risks.....28

8 Revision History.....29

1 Purpose/Scope

- 1.1 Establish minimum standards for all Cleveland-Cliffs Steel Operations to protect employees and contracted workers that are exposed to potential falls greater than 4 feet or working above dangerous equipment. In addition, it establishes the scope, training, and use of fall protection systems by Cleveland-Cliffs Steel employees in Cleveland-Cliffs Steel facilities.
- 1.2 This standard shall establish a means to analyze elevated work tasks, determine appropriate controls and protect personnel from fall to below hazards. All people performing work on Cleveland-Cliffs Steel property shall follow and comply with the requirements in this standard, as it has been developed to meet or exceed applicable OSHA regulatory requirements.

2 Responsibilities and Authorities

2.1 This Standard was jointly developed with participation from the represented workforce and management.



2.2 Cleveland-Cliffs Employees and Contractor Employees shall:

- Notify supervisors or other management personnel when fall exposures exist and are not adequately controlled.
- Attend required training.
- Properly wear all required fall protection equipment when performing a task or activity in an unprotected area where a fall potential of greater than 4 feet exist.
- Inspect fall protection equipment before use to identify any deficiencies.
- Maintain and store personal fall protection equipment following this standard.
- Make personal fall protection components available for annual inspection.

2.3 Supervisors and Management Personnel shall in addition to the above:

- Assess the work area for potential fall hazards, working-at-height scenarios and use the hierarchy of controls to mitigate fall exposures.
- Ensure that only trained and authorized personnel are assigned work that requires fall protection equipment.
- Ensure employees attend required fall protection training.
- Ensure training is documented and records are available.
- Ensure that approved fall protection equipment is available in adequate supply.

2.4 Engineering Department shall:

- Ensure a documented process is in place for engineering, inspecting, and testing of fall protection systems.
- Confirm the qualification and certification of 3rd party inspectors of engineered systems.
- Verify and document structural integrity of roofs and other structures for fall-through protection.

Fall Prevention & Protection Standard

- Maintain documents pertaining to fall protection system design as required for regularly scheduled system recertification.

2.5 Purchasing shall:

- Ensure only approved fall protection equipment is made available that is compliant with regulatory requirements and Cleveland-Cliffs standards.
- Ensure contractors that have the potential for working at heights, understand the requirements stipulated in this Standard, and that contractor personnel have the necessary training to use and wear fall protection equipment.

2.6 Site Safety Department shall:

- Establish and maintain guidance and documents in compliance with this Standard.
- Review and approve the need for new and additional fall protection equipment, including Engineered Systems and components (Gemtor, Railmaster, etc.) before use.
- Audit for compliance with this Standard.
- Ensure Fall Protection Training is made available and is conducted in compliance with this Standard.
- Conduct investigation for all fall to below incidents.
- Evaluate employee needs for special harnesses or other alternative fall protection needs.

3 References and Definitions

3.1 References

- 1910.25
- 1910.28
- 1910.29
- 1910.140
- 1926.500
- 1926.501
- 1926.502
- 1926.503
- ANSI Z359.1 (*The Fall Protection Code*)
- ANSI Z359.14 – 2014 (*Self Retracting Devices*)
- ANSI Z359.14 – 2021 (*Self Retracting Devices*)
- ANSI 359.6 (*Fall Protection & Fall Restraint*)

Δ

Fall Prevention & Protection Standard

3.2 Definitions

Anchorage or Anchorage Point: A secure point of attachment for lifelines, lanyards, or deceleration devices capable of supporting 5,000 pounds per person or is designed by a qualified person and are a part of a complete fall protection system maintaining a safety factor of at least two.

Arresting Force: The force generated by arresting the test weight that is transmitted through the fall arresting system components to the anchorage or load cell.

Authorized Person: A person approved or assigned to perform a specific type of duty or duties or to be at a specific location or job site (e.g., building maintenance, roof repair, etc.). Employees who are required to wear fall protection during work activities.

Body Harness: A design of straps which may be secured about the employee in a manner that will distribute the fall arrest forces over at least the thighs, pelvis, waist, chest, and shoulders with means for attaching it to other components of a personal fall arrest system.

Competent Person: A person who is capable of identifying existing and predictable hazards in any personal fall protection system or any component of it, as well as in their application and uses with related equipment, and who has authorization to take prompt, corrective action to eliminate the identified hazards.

Cage/Basket Guard: An enclosure that is fastened to the side rails of the fixed ladder or to the structure to encircle the climbing space of the ladder for the safety of the person who must climb the ladder.

Connector: A device used to couple (connect) parts of the personal fall arrest system together.

Carabiner: A connector component comprised of a trapezoidal or oval-shaped body with a normally closed gate. The gate, when released, automatically closes to retain the object. All carabiners used in fall protection systems must be the automatic locking type and have a minimum tensile strength of 5,000 pounds (22 kN).

Dangerous Equipment: Equipment (such as pickling or galvanizing tanks, degreasing units, machinery, electrical equipment, and other units) which, as a result of form or function, may be hazardous to employees who fall onto or into such equipment.

Deceleration Device: Any mechanism which serves to lessen the arresting force on the body during a fall.

Deceleration Distance: The vertical distance a falling employee travels from the point at which the deceleration device begins to operate, excluding lifeline elongation and free fall distance, until stopping. It is measured as the distance between the location of an employee's body harness attachment point at the moment of activation (at the onset of fall arrest forces) of the deceleration device during a fall, and the location of that attachment point after the employee comes to a full stop.

Fall Prevention & Protection Standard

Fall Arrest System: A system that prevents a worker from falling and contacting a lower level. It consists of an anchorage, connectors, a full body harness, and may include a lanyard, deceleration device, lifeline, or any combination of the before-mentioned items.

Fall Clearance Calculation: The total calculated distance required to prevent workers from hitting the ground or lower level, in case of a fall.

Free Fall: The act of falling before a personal fall arrest system begins to apply force to arrest the fall.

Free Fall Distance: The vertical displacement of the fall arrest attachment point on the employee's body harness between the onset of the fall and just before the system begins to apply force to arrest the fall. This distance excludes deceleration distance, and lifeline/lanyard elongation, but includes any deceleration device slide distance or self-retracting lifeline/lanyard extension before they operate and fall arrest forces occur. (Always minimize Free Fall Distance not to exceed 6 feet.)

Fixed Ladder: A ladder which is permanently attached to a structure, building or equipment. (Does not include ship stairs, step bolts or manhole steps.) All new or modified fixed ladders greater than 24 feet shall have a ladder safety system or fall arrest system. Existing fixed ladders must have a ladder safety system or fall arrest system by November 2036.

Guardrail System: A barrier erected along an unprotected or exposed side, edge, or other areas of a walking/working surface to prevent workers from falling to a lower level. The system consists of a top rail at 42 inches (plus or minus three inches) above the walking surface, a midrail at a height midway between the top edge of the guardrail and the walking surface. A toe board is required to ensure falling object protection along the exposed edge of overhead walking/working surfaces sufficient to protect employees below.

Handrail System: A barrier erected to prevent employees from falling to lower levels. Handrails are not less than 30 inches and not more than 38 inches measured from the edge of the stair tread. There must be a minimum of 2.5 inches clearance between the handrail and any other object.

Lanyard: A flexible line of rope, cable, or strap with self-locking snap hook connectors at each end for connecting to body harnesses or deceleration devices.

Ladder Safety System: A system (usually consisting of a carrier, safety sleeve, lanyard, connectors, and body harness) attached to a fixed ladder designed to eliminate or reduce the possibility of a worker falling from the ladder.

Leading Edge: The edge of a floor, roof or other walking/working surface which changes location as additional floor, roof, etc., is placed or constructed. A leading edge is considered an unprotected side or edge.

Lifeline: A component consisting of a flexible line for connection to an anchorage at one end to hang vertically (vertical lifelines), or for connection to anchorages at both ends to stretch horizontally (horizontal lifeline), and which serves as a means for connecting other components of a personal fall arrest system to the anchorage.

Fall Prevention & Protection Standard

Low-Slope Roof: A roof having a slope less than or equal to 4 in 12 (vertical to horizontal).

Personal Fall Arrest System: System used to arrest an employee in a fall from a working level. It consists of an anchorage, connectors, full body harness and a self-retractable lifeline/lanyard.

Personal Fall Protection: A system that is used to arrest an employee in a fall from a working level. The fall protection must consist of a full body harness, a lanyard with a deceleration device and an anchorage or lifeline with approved hooking devices as specified in Cleveland-Cliffs Fall Protection Standard.

Qualified Person: One who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, has successfully demonstrated ability to solve or resolve problems relating to the subject matter, the work or the project.

Rope Grab: A cam-type deceleration device that travels on a lifeline and automatically engages the lifeline and locks to arrest an employee's fall.

Safety Factor: An added margin of security against risk.

Safety Net Systems: A horizontal or semi-horizontal, cantilever-style barrier that uses a netting system to stop falling workers before they contact a lower level or obstruction.

Safety Monitor: A competent person responsible for recognizing and warning workers of fall hazards.

Self-Retracting Lifeline/Lanyard: Deceleration device containing a drum-wound line which can be slowly extracted from, or retracted onto, the drum under slight tension during normal employee movement, and which after the onset of a fall, automatically locks the drum and arrests the fall.

Steep Roof: Roof having slope of greater than 4 in 12 (vertical to horizontal).

Snap Hook: A connector comprised of a hook-shaped member with a closed keeper, which may be opened to permit the hook to receive an object, and when released, automatically closes to retain the object. Snap hooks must be self-closing with a self-locking keeper, which remains closed and locked until unlocked and pressed open for connection or disconnection, thus preventing the opportunity for the object to "roll out" of the snap hook. Snap hooks used for fall protection systems must have a 3,600-pound gate strength.

Swing Fall: The potential for a worker using a personal fall arrest system to fall and swing in a pendulum motion, subjecting the worker to strike a fixed object or other hazards. This can occur if a worker falls and the anchorage is not directly overhead, and a fall allows the worker to swing toward the anchor point.

Tie Off: The act of a person wearing personal fall protection equipment, connecting directly to an anchorage point. 100 % tie off is required when exposed to a fall hazard while using Travel Restraint or Fall Arrest systems.

Fall Prevention & Protection Standard

Toe Board: A solid horizontal guard installed along the exposed edge of overhead walking/working surfaces to prevent the fall of materials and equipment to lower levels and protect employees below. They must be a minimum of 3.5 inches and not have an opening more than 0.25 inches above the walking/working surface.



Total Fall Distance: Adding the free fall distance, lanyard deceleration distance *depending on the equipment (24" for ANSI Z359.14-2014 Class A, 54" for ANSI Z359.14-2014 Class B, 42" for ANSI Z359.14-2021 Class 1 and Class 2)*, harness stretch, the distance your body extended from the dorsal ring of your harness attached to your anchorage point, *and a 3-foot safety factor*.

Trauma Straps: A pair of straps attached to a full body harness that allows a worker, who is suspended, to stand up in their harness and to relieve the pressure being applied to the arteries and veins around the top of the legs.

Travel Restraint (Tether) Line: A rope or wire rope used to transfer forces from a body support to an anchorage or anchorage connector in a travel restraint system.

Travel Restraint System: A combination of an anchorage, anchorage connector, lanyard (or other means of connection) and body support intended to be used by an employee to limit travel to prevent exposure to a fall hazard. A travel restraint system is used such that it does not support any portion of the employee's weight; otherwise, the system would be a positioning system or a personal fall arrest system. Travel restraint lines must be capable of sustaining a tensile load of at least 5,000 pounds (22.2 kN).

Unprotected Sides or Edges: Any side or edge (except at entrances to points of access) of a walking/working surface; e.g., floor, roof, ramp, or runway where there is no wall or guardrail system at least 39 inches high.

Walking/Working Surface: Any surface, whether horizontal or vertical, on which an employee walks or works, including but not limited to floors, roofs, ramps, bridges, runways, framework, and concrete reinforcing steel.

Work Positioning: A system consisting of a body harness, lanyards, and anchorage devices rigged to allow a person to be supported on an elevated vertical surface, such as a wall or fixed ladder, and work with both hands free. Attachments should be at the workers hips as preferred method while using a work positioning system.

4 Resources

- 4.1 Training is required for authorized persons and contractors who work in areas where fall hazards exist and fall protection equipment is required.
- 4.2 Fall prevention, travel limiting and fall arrest equipment must be purchased and provided to employees who are exposed to falls greater than 4 feet or working above dangerous equipment.
- 4.3 Fall protection equipment will be inspected as appropriate.

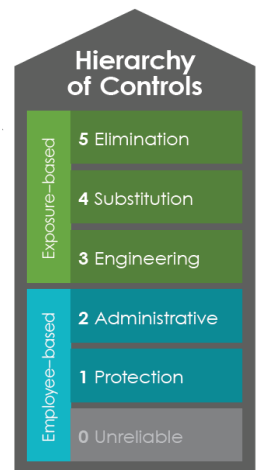
5 Requirements

5.1 Application

- 5.1.1 Cleveland-Cliffs Steel employees, contractors, sub-contractors, service providers, vendors and visitors when working at any elevation greater than four (4) feet from the floor, deck, platform, or ground level that is not adequately protected on all sides by standard handrails or guardrails shall be required to use personal fall protection.
- 5.1.2 Cleveland-Cliffs Steel employees, contractors, sub-contractors, service providers, vendors, and visitors when at any elevation above dangerous equipment shall be protected from falling into or onto the dangerous equipment by guardrails, equipment guards or appropriate fall protection systems.
- 5.1.3 Cleveland-Cliffs Steel employees, contractors, sub-contractors, service providers, vendors and visitors working in an aerial lift, scissors lift, on scaffolding without standard handrail or exposed to a fall hazard around an excavation (4 feet or deeper) are also required to use personal fall protection.
- 5.1.4 Every effort must be made by all workers to eliminate falls to a lower level or into or onto dangerous equipment.
- 5.1.5 Use the Take 5 process to assess the job and use the Hierarchy of Controls when controlling fall to below hazards.

Examples:

- Eliminate the need for elevated work by moving valves to floor level.
- Guard from falls by installing permanent standard railing.
- Use scaffolding with handrails, mid-rails, and toe boards.
- Use aerial lifts.
- Use travel restraint when other exposure-based controls are not available.
- Use fall arrest systems when the work doesn't allow for higher-level controls.
- Safety Net System as the last line of protection when no other can be achieved.



Fall Prevention & Protection Standard

- 5.1.6 Distance alone from a fall to below hazard does not afford adequate protection.
- 5.1.7 All roofs at Cleveland-Cliffs Steel require fall off and fall through protection. Refer to the plant's specific roof access procedure. Fall off protection must be used anytime an employee is not protected by standard walkway and handrails or other engineered system. Barricade and restrict access to roofs or other working surfaces where structural integrity concerns exist or where known fall hazards exist.
- 5.1.8 Travel Restraint devices are used to prevent employees from approaching an area where he or she may fall, such as in aerial and scissors lifts and other personal lifting devices. Only full-body harnesses are permissible as Travel Restraint devices. Lanyards used with the harness must have double locking snap hooks or double locking carabineers.
- 5.1.9 Travel Restraint systems shall require:
- Full body harness equipped with trauma straps.
 - Non-shock absorbing lanyard.
 - Lanyard must be of a length that will not allow access to the fall hazard.
 - Anchorage capable of supporting at least 5,000 pounds per user or designed by a qualified person and are a part of a complete fall protection system maintaining a safety factor of at least two.
 - Any required welding on anchorage must be conducted by a person certified for the specific application and engineering specifications. The person must be certified for the welds required. Contact plant engineering for welding or attachment requirements prior to installation.
- 5.1.10 Personal fall arrest systems shall require: ABC
- A. Anchorage Point/Anchorage Connector
- Anchorage capable of supporting at least 5,000 pounds per user or designed by a qualified person and are a part of a complete fall protection system maintaining a safety factor of at least two.
 - Any required welding on anchorage must be conducted by a person certified for the specific application and engineering specifications. The person must be certified for the welds required. Contact plant engineering for welding or attachment requirements prior to installation.
 - Anchorage is to be installed directly overhead when possible.
 - Workers should not walk more than six (6) feet away from anchorage when possible.
- B. Full body harness equipped with trauma straps.
- C. Connecting Device.

Fall Prevention & Protection Standard



- Self-retracting lanyard (SRL).
- When leading edge or sharp edge potential exists that can damage lanyards during a fall a Leading Edge SRL meeting the requirements of ANSI Z359.14 will be required.
- Connection directly to the D-ring in the middle of the back, *unless the manufacturer provides an approved connector (e.g., Dual Leg Personal SRL).*
- No additional lanyard or other lengthening devices (Short D-ring extenders are allowed if included in the design and any calculations where appropriate).
- Any exception must be reviewed and approved in writing by the Site Safety Manager. Site Safety Department will review and approve all new and additional fall protection equipment, including Engineered Systems and components (Gemtor, Railmaster, etc.) prior to use.

5.1.11 Self-retracting lifelines/lanyard must be capable of sustaining a minimum tensile load of 5,000 pounds (22.2 kN) applied to the device with the lifeline or lanyard in the fully extended position.

5.1.12 D-rings and snap-hooks must be capable of sustaining a minimum tensile load of 5,000 pounds (22.2 kN).

5.1.13 All self-retractable lifeline/lanyard and connecting means shall have double locking snap hooks.



5.1.14 All lanyard, deceleration device, and full body harness connecting means shall be made of drop forged or pressed steel and cadmium plated, *or other ANSI approved and tested materials.* The surfaces shall be smooth and free of sharp edges.

Note: Identify and document all electrical fall hazards. If an arc-rated harness is used when working near energized electrical sources, refer to the arc-rated harness manufacturers' guidelines regarding recommended lanyard and connecting means.

5.1.15 All self-retractable lifeline/lanyard, full body harnesses and associated hardware shall be inspected before and after each use.

5.1.16 The anchorage (tie-off point) shall be capable of supporting at least 5,000 pounds per employee attached. If there is any doubt about the strength of the anchorage point, DO NOT TIE OFF; find an alternate anchorage point.

5.1.17 Employee and tools over 310 pounds: Ensure the combined weight (tools and body) does not exceed the limits of the equipment. If the system is used by an employee having a combined weight (tools and body weight) of 310 pounds or more, special personal fall arrest equipment will need to be ordered to accommodate the individual. Please contact the Safety Department for proper equipment, training and job set up if a special harness is needed.

Fall Prevention & Protection Standard

- 5.1.18 If any part of a fall protection system is subjected to a free fall, all components shall be taken out of service immediately and given to the Safety Department for discussion with competent persons and disposition.
- 5.1.19 If any employee experiences a free fall, the Safety Department shall be notified immediately.

5.2 Engineered Systems

- 5.2.1 Lifeline systems must be engineered to have appropriate anchorages, the strength of line designed to hold a specific number of individuals connected to it, line strength to aid in the arrest of a fall, and durability to hold a fallen employee(s) suspended until a rescue can occur. Only a Qualified Person may design a fall protection system or certify an existing system or anchorage point.
- 5.2.2 Engineered Systems and components (Gemtor, Railmaster, etc.) must be reviewed by a competent person prior to the Safety Department's approval to purchase and/or use. A competent person reviews the installation as a part of the fall protection system.

5.3 Tie off Procedures

- 5.3.1 When using a full-body harness for fall protection, connecting devices shall be attached to the D-ring in the middle of the back, *unless the manufacturer provides an approved connector (e.g., Dual Leg Personal SRL)*.
- 5.3.2 Snap hooks, including pelican hooks and carabiners, shall be of a design, size, and be compatible with the member that they are attached in order to prevent unintentional disengagement by depression of the keeper or side loading the keeper beyond its rating.
- 5.3.3 Side D-rings are for travel restraint or work positioning only and shall never be used for fall arrest connections. Whenever possible, tie off to an anchorage point at or above the D-ring of the harness.
- 5.3.4 For fall arrest systems on ladder safety systems, a front D-ring will be used or a back D-ring will be used based on the system design.
- 5.3.5 All new or modified fixed ladders that extend more than 24 feet shall have a ladder safety system or fall arrest system. Existing fixed ladders must have ladder safety system or fall arrest system by November 2036.
- 5.3.6 Workers shall tie off in a manner that ensures no lower level is struck during a free fall.

Fall Prevention & Protection Standard

5.3.7 Completely understand Fall Clearance Distance. This is the total distance for the fall arrest equipment to stop a fall plus a 3-foot safety factor. Always consider the following when determining fall protection equipment and proper tie-off location:

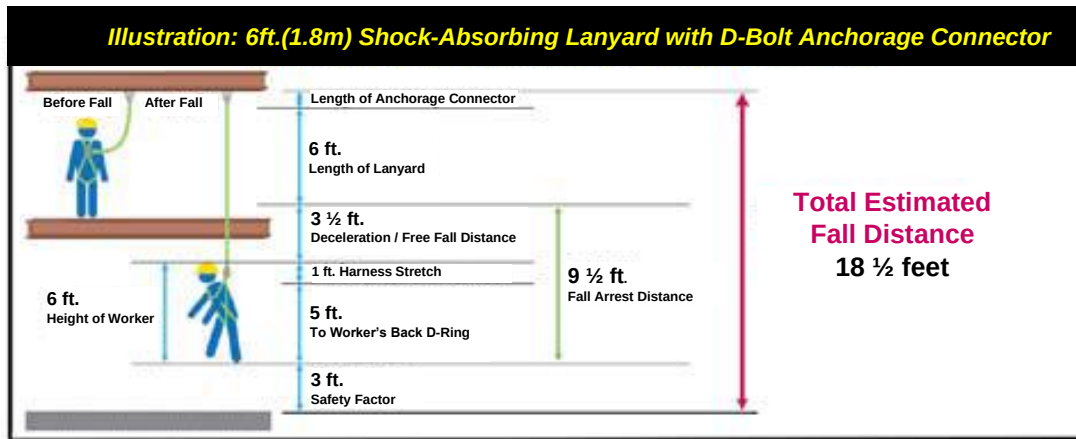
- Location and height of the anchorage.
- Free Fall Distance.
- The maximum arrest or deceleration distance, *depending on the equipment (24" for ANSI Z359.14-2014 Class A, 54" for ANSI Z359.14-2014 Class B, 42" for ANSI Z359.14-2021 Class 1 and Class 2). Class 1 must be attached to an anchor at or above the D-ring.*
- The distance that any deployable device will expand.
- Stretch of harness and other components.
- The height of the employee.
- Swing Fall or deflection of non-rigid anchors such as horizontal lifelines.
- Safety factor of 3 feet.
- Workers should not walk more than six (6) feet away from anchorage, when possible, to minimize free fall distance and swing fall.
- Add length of D-ring Extension if applicable.



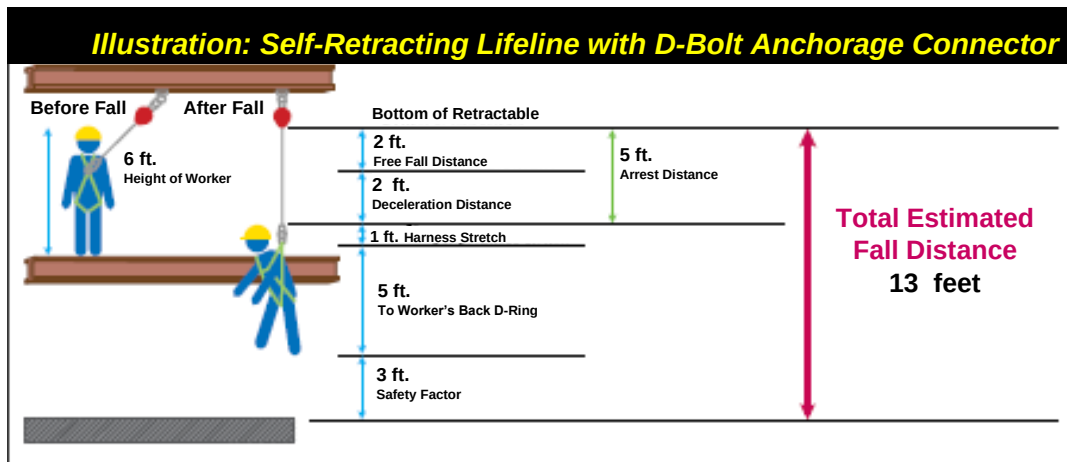
Fall Prevention & Protection Standard

- 5.3.8 When calculating actual distance, make necessary adjustments based on job set-up and equipment used.

EXAMPLE



EXAMPLE



- 5.3.9 Anchorage points must, where practical, be above the head of the worker, reduce possible free fall distance, prevent swing fall hazards, and provide clear space in the potential fall path to avoid striking an object.
- 5.3.10 In considering the most desirable anchorage for tie-off, any sharp object that the lanyard may contact while at the work location or during a fall must be avoided. Lanyards must never be tied off to shafts or other parts of machinery which are subject to movement or rotation.

Fall Prevention & Protection Standard

- 5.3.11 Use caution while wearing a body harness and lanyard when you are not tied off. Wear the lanyard in such a manner that it will not get caught on or in equipment, structures or moving machinery. No part of the lanyard shall be allowed to dangle freely when not tied off. The lanyard shall be secured tightly against the user's body.
- All adjustment straps of the body harness shall be kept snug against the body.
- 5.3.12 Where movement beyond the reach of a single lanyard (including SRLs) is required, a hazard assessment must be conducted to determine the need for lifelines and dual SRLs with twin-leg lanyards. Where dual SRLs with twin-leg lanyards are required, the person must ensure that at least one leg is attached all the time insuring 100% tie-off.
- 5.3.13 Self-retractable lifeline/lanyards shall never be wrapped back into themselves unless designed to do so. Wrapping can weaken the lanyard by 75 percent. Instead, beam wraps and other engineered devices shall be used.
- 5.3.14 The use of beam straps, beam clamps, or other temporary tie-off points must be used exclusively as part of a fall protection system and not used for rigging, lifting or other purposes. All anchor points must meet at least 5,000-pound tensile strength requirements and be designed and used for the intended purpose.

EXAMPLES



Fall Prevention & Protection Standard

5.3.15 Transitioning from Aerial Lift/Platform

- If transition from the aerial platform to another location must be made, a hazard assessment shall be completed to ensure that the fall hazard has either been eliminated, or 100% fall protection tie off (twin leg lanyard) is used.
- When transitioning, one leg of the employee's twin-leg (or Y) lanyard must be attached to an approved connection point in the work area being entered. The other leg of the lanyard must be tied off to the anchorage point of the aerial lift.
- The same procedure must be followed when re-entering the aerial lift from the work location.
- Workers in an aerial lift or scissors lift will not wear fall arrest systems unless the lift and the attachment points have been specifically designed to meet all requirements of a fall arrest anchoring system and the requirements of this document. The fall protection system must be designed to comply with the anchorage provided by the manufacturer and appropriate use.

5.4 Crane Runway Fall Protection

5.4.1 The fall arrest anchor system must be designed by a qualified person from either the Engineering Department or outside engineering firm.

5.4.2 Whenever possible, the cable height must be set high enough above crane runway girder to ensure clearances and limit fall arrest distance. In addition, where possible, the cable is to be over entire length of both sides of all crane runways.

5.4.3 When using horizontal lifelines for fall protection on crane runways, the employee shall make sure the lifeline is certified and inspected. All horizontal lifelines must be recertified every five years and illegible tags must be replaced at that time. Approved horizontal lifelines will be identified with an orange metal tag with the following information:

- Building Number
- Crane Number
- Date Installed
- Installation Ref. Drawing Number
- Inspection Ref. Drawing Number
- Maximum Number of Users per Bay
- Maximum Lanyard Length
- Installation PO/WO Number

Fall Prevention & Protection Standard

- 5.4.4 This certification/recertification tag and annual inspection tag (with date and inspector's name) is to be installed at each end of the horizontal lifeline and at each location between the ends where permanent access to the horizontal lifeline exists. Paper copies of all tags are to be kept on file by the Maintenance Section Manager in each department.
- 5.4.5 If a crane runway is not equipped with a personal fall arrest system or temporary lifelines designed by the Engineering Department or outside engineering firm; walking on the crane runway is prohibited. The use of Railmaster or similar devices is allowed when necessary.
- 5.4.6 All persons walking along the crane runway are to have full body harnesses, equipped *with twin-leg SRLs, or Travel Restraint System*, when needed, based on the design of the system to achieve 100% tie off. Dual-leg lanyards are necessary to meet OSHA standards when the person comes to a column and must hook the second leg to the cable on the opposite side of the column before unhooking the first leg.

Δ

5.5 Handrail and Guardrail Systems

- 5.5.1 Handrails and guardrails must be made from steel unless the environment dictates the material requires protection from corrosion. Existing wood, chain and wire rope protection shall be replaced. Wood, chain, and wire rope may be used for barriers only. The minimum clearance between the handrail and any other object is 2.5 inches. Where physically possible, access gates used for fall protection should be swing gates designed to not open in the direction of the fall exposure.
- 5.5.2 Guardrail systems must have a rail at 42 inches (plus or minus 3 inches) above the walking surface, a midrail at a height midway between the top edge of the guardrail and the walking surface, installed so that openings do not exceed 19 inches at its least dimension. A minimum 3.5-inch toe board must be installed along the exposed edge of overhead walking-working surfaces to protect employees below. For stair rail systems, built before January 17, 2017, the top rail may serve as a handrail, provided the top rail is 36 to 38 inches in height. For stair rail systems installed on or after January 17, 2017, the top rail and handrail must be separate. The top rail must be at least 42 inches in height and the handrail must be 30 to 38 inches in height. Guardrails and handrails must withstand 200 pounds of force in any direction.
- 5.5.3 These handrails and guardrails will be placed in the following areas if necessary or feasible based on job location or requirements:

Fall Prevention & Protection Standard

- On all open sided floors.
- Elevated platforms, walkways, and other elevated walking/working surfaces.
- Tops of vessels, tanks and other structures or equipment that could be accessed by personnel.
- Around all open excavations or pits over 4 feet deep.
- On edges of roofs to provide fall off protection when the roof is structurally sound.

5.6 Inspection

- 5.6.1 Each plant will establish a process for full body harnesses, lanyards, and their accessories to be inventoried.
- 5.6.2 Personal fall arrest systems shall be inspected prior to each use for wear, damage, and other deterioration. Defective components shall be removed from service and destroyed. The employee shall inspect all fall protection equipment for damage or excessive wear.
- 5.6.3 Full Body Harness Inspection
- 5.6.3.1 Harness straps are to be inspected by beginning at one end, holding the body side of the harness towards you, grasp the harness with your hands 6 to 8 inches apart. Bend the harness strap in an inverted "U". The surface tension resulting makes damaged fibers or cuts easier to see. Follow this process the entire length of the harness. Watch for frayed edges, broken fibers, pulled stitches, cuts or chemical/burn damage.
- 5.6.3.2 Check the D-ring and D-ring metal wear pad (if any) for distortion, cracks, breaks and rough or sharp edges.
- 5.6.3.3 Attachment of buckles and D-rings should be given special attention. Note any unusual wear, frayed, cut fibers or distortion of the buckles or D-rings. Rivets must be tight and unmovable with fingers. The body side rivet base and outer rivet burr must be flat against the material. Bent rivets will fail under stress.
- 5.6.3.4 Inspect for frayed or broken strands. Broken webbing strands generally appear as tufts on the webbing surface. Any broken, cut or burned stitches will be readily seen.
- 5.6.3.5 The tongue or billets of the leg straps receive heavy wear from repeated buckling and unbuckling. Inspect for loose, distorted, or broken grommets. Straps must never have additional punched holes.

Fall Prevention & Protection Standard

5.6.3.6 Buckle tongues must be free of distortion in shape and motion. They must overlap buckle frames and move freely back and forth in their socket. The roller must turn freely on the frame. Check for distortion or sharp edges.

5.6.3.7 Trauma straps must be present and not damaged.

5.6.3.8 Any additional inspection procedures required by the equipment manufacturer as provided in training.

5.6.4 Lanyard Inspection.

5.6.4.1 Travel Restraint Lanyard Inspection: When inspecting lanyards, begin at one end and work to the opposite end. Slowly rotate the lanyard so that the entire circumference is checked. Stitched (spliced) ends require extra attention.

5.6.4.2 The double locking snaps will be inspected closely for hook and eye distortions, cracks, corrosion, or pitted surfaces. The keeper (latch) must seat into the nose without binding and not be distorted or obstructed. The keeper spring will exert sufficient force to firmly close the keeper. Keeper locks must prevent the keeper from opening when the keeper closes.

5.6.4.3 The thimble must be firmly seated in the eye of the splice, and the splice must have no loose or cut strands. In addition, the edges of the thimble must be free of sharp edges, distortions, or cracks.

5.6.4.4 While bending a web lanyard over a pipe or mandrel, observe each side of the webbing. This will reveal any cuts or breaks, swelling, discoloration, cracks, charring or obvious signs of chemical or heat damage.

5.6.4.5 Rotation of the rope lanyard while inspecting from end to end will reveal any fuzzy, worn, broken, or cut fibers. Weakened areas from extreme loads will appear as a noticeable change in original diameter. The rope diameter must be uniform throughout, following a short break-in period.

5.6.4.6 Any additional inspection procedures required by the equipment manufacturer as provided in training.

5.6.5 Snap hooks (including carabineers).

5.6.5.1 Snap hooks must be of the locking type designed and used to prevent disengagement of the snap hook by the contact of the snap hook keeper by the connected member. Snap hooks must be proof tested by the manufacturer to a minimum tensile load of 3,600 pounds, (normal 5,000 pounds) without cracking, breaking, or taking permanent deformation.

Fall Prevention & Protection Standard

5.6.5.2 The keeper of the snap hook must have a spring tension that will not allow the keeper to open with a weight of 25 pounds or less, but the keeper must begin to open with a weight of 4 pounds supported on the keeper against the end of the nose.

5.6.5.3 Inspection of the snap hook must include:

- Signs of cracking.
- Sharp edges.
- Signs of breaking.
- Permanent deformation.
- Pitted or corroded surfaces.
- Keeper must seat properly without binding or obstruction.
- Keeper spring must exert sufficient force to ensure closure.
- Locks must prevent keeper from opening.

5.6.6 Inspection of Retractable Lanyards:

- Anchorage connectors shall be checked for bent, cracked, or distorted components.
- With the device in the mounted position, test the lanyard retraction and tension by pulling out several feet of the lanyard and allowing it to retract back into the unit. Always maintain a light tension on the lanyard as it retracts. The lanyard should pull out freely and retract all the way back into the unit.
 - a. If the lanyard does not pull out smoothly or sticks when retracting, pull the entire lanyard out of the housing and allow it to retract slowly under tension.
 - b. The lanyard must be checked for signs of damage. Inspect for cuts, burns, corrosion, kinks, frays or worn areas. Inspect any sewing for loose, broken, or damaged stitches.
- The braking mechanism shall be tested by grasping the lifeline above the impact indicator (if equipped) and applying a sharp, steady pull downward which will engage the brakes. There should be no slippage of the lifeline while the brakes are engaged. Once tension is released, the brakes will disengage, and the unit will return to the retractable mode.
- Check the snap hook to ensure that it operates freely and locks. Inspect the snap hook for any signs of damage to the keepers and any bent, cracked, or distorted components.
- Load indicators, if equipped, will typically be located above the snap hook. An indicator will be exposed when subjected to fall arresting forces. Check manufacturer instructions for indicator type and location.

Fall Prevention & Protection Standard

- Any additional inspection procedures required by the equipment manufacturer shall be observed.

5.6.7 Visual Indications of Damage to Webbing and Lanyards:

5.6.7.1 Nylon & Cordura:

- Heat Damage – In excessive heat, nylon becomes brittle and has a shriveled brownish appearance. In addition, fibers will break when flexed. Therefore, nylon must never be used above 200 degrees Fahrenheit. *Refer to limitations listed in the manufacturer's user manual.*
- Chemical Damage – Color change usually appears as a brownish smear or smudge. Transverse cracks when harness or lanyard is bent over a pipe. There will also be a loss of elasticity in the harness or lanyard.
- Flame/Molten Metal Damage – Webbing strands fuse together. Hard, shiny spots and brittle feel but will not support combustion.
- Paint/Solvent Damage – Paint, which penetrates and dries, restricts the movement of fibers. Drying agents and solvents in some paints will appear as chemical damage.

5.6.7.2 Polyester & Dacron:

- Heat Damage – Same as nylon above, except do not use above 180 degrees Fahrenheit. *Refer to limitations listed in the manufacturer's user manual.*
- Chemical Damage – Same as nylon above.
- Flame/Molten Metal Damage – Same as nylon above, except will support combustion.
- Paint/Solvent Damage – Same as nylon above.

5.6.7.3 Cable:

- Slightly bend and rotate the cable.
- Visually check for damaged, kinked, broken strands or unusual discoloration or wear patterns.

5.7 **Cleaning and Storage of Fall Protection Equipment**

- 5.7.1 Cleaning: Wipe off all surface dirt with a sponge dampened in plain water. Squeeze the sponge dry. Dip the sponge in a mild solution of water and commercial soap or detergent. Work up a thick lather with vigorous back and forth motions. Wipe the belt dry with a clean cloth. Equipment shall hang freely to dry thoroughly without close exposure to heat, steam, or prolonged periods of sunlight.

Fall Prevention & Protection Standard

- 5.7.2 Never use chemical cleaners such as naphtha, turpentine, acetone, or other similar products. Never use bleach, rubbing alcohol, hydrogen peroxide or other products with sanitizing or disinfecting claims. The effect of these substances on equipment has not been evaluated and may compromise material integrity.
- 5.7.3 Storage: Never store personal fall arrest equipment on the ground or outdoors exposed to the elements (e.g., sun, rain, snow), or in an area with exposure to harmful substances or corrosive elements.
- 5.7.4 Refer to the manufacturer's recommendations for additional cleaning and storage considerations.

5.8 Training

- 5.8.1 Each employee authorized to wear fall protection equipment shall be trained to recognize the hazards of falling, and procedures to be followed to minimize these hazards. The employee shall be trained by a person who is competent in the following areas:
- Nature of fall hazards in the work area and how to recognize them.
 - The correct procedures for installing, inspecting, operating, maintaining, and disassembling the personal fall protection systems.
 - The correct use of personal fall protection systems and equipment including, but not limited to, proper hook-up, anchoring and tie-off techniques, and methods of equipment inspection and storage, as specified by the manufacturer.
 - The use and operation of guardrail systems, personal fall arrest systems and other protection to be used.
 - Limitations of fall protection system including proper hook-up, anchoring and tie-off techniques.
 - Requirements set forth in this program and the procedures to be followed to minimize fall hazards.
 - Emergency and Rescue procedures.
- 5.8.2 Content of Fall Protection Training
- OSHA's requirements for fall protection.
 - Contents of Cleveland-Cliffs Steel Fall Protection Standard.
 - Basic skills in fall hazard recognition and control.
 - Donning and Doffing.
 - Proper methods of inspecting personal fall protection equipment.
 - Fall Clearance Calculation methods.
 - Fall protection systems used at Cleveland-Cliffs Steel.
 - Hazards associated with fall protection.
 - Rescue and retrieval requirements.

Fall Prevention & Protection Standard

- 5.8.3 Training must occur initially before exposure, annually, and when there are changes in the types of fall protection systems or equipment or there are changes in the workplace that render previous training obsolete.
- 5.8.4 The employee shall be retrained when there is reason to believe the employee does not have the understanding and skill required to follow this procedure. A competent person must conduct any required retraining.
- 5.8.5 The training shall be documented with the name of the employee, identification number, the date(s) of training and the signature of the person conducting the training.
- 5.8.6 After the completing the training, a documented quiz or test must be administered to ensure that the employee has acquired the understanding, knowledge and skills necessary to safely perform their duties.

5.9 Safety Net Systems

- 5.9.1 Safety Net Systems can be used when no other fall protection system is available. Safety Nets must be designed and tested by a qualified person and inspected by a competent person.
- 5.9.2 A safety net system can be erected in a vertical or horizontal plane or both. It must protect the worker from free falling to a lower level or structure. In addition, it must protect employees from falling tools or debris from the worksite.
- 5.9.3 Safety nets must be installed as close to the area as possible and must never be more than 30 feet below the highest working surface from which a person may fall.
- 5.9.4 Nets must be hung with sufficient clearance under them to prevent contact with the surface or structures below when subjected to an impact force equal to the drop test specified by OSHA.
- 5.9.5 Safety nets and safety net installations must be drop tested at the job site after initial installation before using as a fall protection system, whenever relocated, after major repair, and at six-month intervals, if left in one place.
- 5.9.6 If it can be demonstrated that it is unreasonable to perform the drop test, a competent person must certify the net and that the net installation is in compliance.
- 5.9.7 The mesh size of nets must not exceed 6 inches by 6 inches. All new nets must meet accepted performance standards of minimum impact resistance as determined and certified by the manufacturer and must bear a label of proof testing. Border ropes must provide a minimum breaking strength of 5,000 pounds.

Fall Prevention & Protection Standard

- 5.9.8 Safety nets must not extend outward from the outermost projection of the work surface as described by the OSHA table. Forged steel safety hooks or shackles must be used to fasten the net to its supports. Connections between net panels must develop the full strength of the net.
- 5.9.9 Net systems must be checked and documented daily and cleaned whenever debris is in the net. Remove any damaged nets for repair or replacement. When nets are removed, no work must take place in that area unless another fall protection system is used. Spare nets may be installed to replace damaged nets if all other components pass an inspection, and a competent person checks the new net.

5.10 Rescue

- 5.10.1 Emergency and rescue procedures must be developed for rescuing a worker within 10 minutes who has fallen from elevated heights.
- 5.10.2 An Emergency and Rescue Plan must be developed by the operating department in coordination with the Rescue Team. It must include an effective rescue summoning method. It must be in place and discussed prior to any Personal Fall Protection System use.
- 5.10.3 Emergency teams must be trained on the procedure.

5.11 Investigation of a Fall

- 5.11.1 All employees involved in a fall arrest or fall must be sent immediately for a medical evaluation to determine the extent of injuries.
- 5.11.2 All falls must be investigated following Cleveland-Cliffs Steel incident investigation procedures. A competent person in fall protection must attend the investigation as a resource.
- 5.11.3 Once exposed to a fall, remove fall protection equipment from service immediately and destroy when required after analysis and review by representatives from the Safety and Engineering Departments. Remove the destroyed equipment from any inventory list.

6 Forms and Documented Information

- 6.1 All training must be documented.
- 6.2 All investigations must be documented in the appropriate Cleveland-Cliffs Steel investigation format.
- 6.3 Certified anchorage points and lifelines (engineered systems) in buildings, on roofs and external working places where appropriate must be inventoried, inspected and documented.
- 6.4 Annual inspection of personal fall protection equipment must be documented.

7 Opportunities and Risks

- 7.1 This standard provides a consistent approach to addressing fall hazards at all Cleveland-Cliffs Steel facilities.
- 7.2 This standard provides opportunities for employees and contractors to be protected from falls to below and falling into hazardous equipment.
- 7.3 Risks addressed in this standard:
- Working at elevated heights.
 - Proper design and use of fall protection equipment.
 - Handrail and guardrail design.
 - Protection of workers from falling objects.
 - Appropriate training for workers at risk.
 - Proper inspection, cleaning, and storage of fall protection equipment.

8 Revision History

Rev.	Date	Summary of change	Authorized by
1	9-1-2022	Initial issue	K. D. Berard
2	11-3-2023	Changes are identified with a delta.  - Added Reference to ANSI Z359.14 – 2014 and ANSI Z359.14 – 2021 (Self Retracting Devices). - Updated the definition of Total Fall Distance to be consistent with Class 1 and Class 2 SRL deceleration distance. - Personal fall arrest systems with connection directly to the D-ring in the middle of the back, unless the manufacturer provides an approved connector (e.g., Dual Leg Personal SRL). - Allowance for materials other than cadmium plated steel for connecting means if meeting ANSI approved, tested materials. - Added option for Travel Restraint System for Crane Runway Fall Protection.	Nathan Berry